

**EDUGUIDE AI
SMART CAREER & OPPORTUNITY FINDER**

A PROJECT REPORT

Submitted by

NALINAA AC (8115U24AM040)

in partial fulfilment for the award of the course

AMB 1221-MACHINE LEARNING FUNDAMENTALS

IN

**DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)**



**K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY**



**ANNA UNIVERSITY
CHENNAI 600 025**

MAY 2026

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Under the Guidance of

Mrs. C.MUTHUSELVI

Department of Artificial Intelligence and Data Science

K. RAMAKRISHNAN COLLEGE OF ENGINEERING



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BONAFIDE CERTIFICATE

Certified that this project report titled **“EDUGUIDE AI SMART CAREER & OPPORTUNITY FINDER”** is the Bonafide work of **NALINAA AC (8115U24AM040)** who carried out the work under my supervision.

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DECLARATION BY THE CANDIDATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva-Voice held at K. Ramakrishnan College of Engineering on _____

SIGNATURE OF THE CANDIDATE

ACKNOWLEDGEMENT

I thank the almighty GOD, without whom it would not have been possible for me to complete my project.

I wish to address my profound gratitude to **Dr.K.RAMAKRISHNAN**, Chairman, K. Ramakrishnan College of Engineering(Autonomous), who encouraged and gave me all help throughout the course.

I extend my hearty gratitude and thanks to my honorable and grateful Executive Director **Dr.S.KUPPUSAMY, B.Sc., MBA., Ph.D.**, K. Ramakrishnan College of Engineering(Autonomous).

I am glad to thank my Principal **Dr.D.SRINIVASAN, M.E., Ph.D.,FIE., MIIW., MISTE., MISAE., C.Engg**, for giving me permission to carry out this project.

I wish to convey my sincere thanks to **Dr.B.KIRAN BALA, M.E., M.B.A., Ph.D.**, Head of the Department, Artificial Intelligence and Data Science for giving me constant encouragement and advice throughout the course.

I am grateful to **Mrs. C.MUTHUSELVI, M.E.**, Assistant Professor, Artificial Intelligence and Data Science, K. Ramakrishnan College of Engineering (Autonomous), for her guidance and valuable suggestions during the course of study.

Finally, I sincerely acknowledged in no less terms all my staff members, my parents and, friends for their co-operation and help at various stages of this project work.

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K. RAMAKRISHNAN COLLEGE OF ENGINEERING

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ANNA UNIVERSITY, CHENNAI



DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

VISION

To become a renowned hub for AIML technologies to producing highly talented globally recognizable technocrats to meet industrial needs and societal expectation.

MISSION

Mission of the Department

- M1** To impart advanced education in AI and Machine Learning, built upon a foundation in Computer Science and Engineering.
- M2** To foster Experiential learning equips students with engineering skills to tackle real-world problems.
- M3** To promote collaborative innovation in AI, machine learning, and related research and development with industries.
- M4** To provide an enjoyable environment for pursuing excellence while upholding strong personal and professional values and ethics.

PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

- PEO1** Excel in technical abilities to build intelligent systems in the fields of AI & ML in order to find new opportunities.
- PEO2** Embrace new technology to solve real-world problems, whether alone or as a team, while prioritizing ethics and societal benefits.
- PEO3** Accept lifelong learning to expand future opportunities in research and product development.

PROGRAM SPECIFIC OUTCOMES (PSO's)

- PSO1** Expertise in tailoring ML algorithms and models to excel in designated applications and fields.
- PSO2** Ability to conduct research, contributing to machine learning advancements and innovations that tackle emerging societal challenges.

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

PO1 Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6 The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO8 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

PO9 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11 Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

EduGuide AI is an intelligent Machine Learning–based recommendation system developed to assist students in identifying suitable internships, skill-development programs, free online courses, and government training opportunities. In the current academic environment, students often struggle to navigate the large number of platforms offering career-related resources, resulting in poor choices, wasted time, and missed opportunities. This project addresses these challenges by collecting student information such as educational background, areas of interest, and existing skills, and analyzing this data using machine learning techniques to generate personalized and relevant recommendations. The system is developed as a web-based application using Python and the Flask framework, ensuring easy accessibility and a user-friendly interface for students with varying levels of technical knowledge. It incorporates modules such as user profile management, interest and skill input, recommendation generation, and course and internship display. The core recommendation engine applies machine learning algorithms to match student profiles with the most appropriate opportunities available. This eliminates manual searching, reduces confusion, and ensures that students receive guidance tailored to their specific needs. EduGuide AI aims to simplify career planning, enhance student employability, and support the development of stronger academic and professional profiles through intelligent, data-driven guidance. The system performs efficiently with minimal computational requirements and is designed to be scalable for future enhancements. Overall, the project demonstrates the practical application of artificial intelligence in the domain of education and career development, and has strong potential for further expansion.

ABSTRACT WITH POs AND PSOs MAPPING

ABSTRACT	POs MAPPED	PSOs MAPPED
The EDUGUIDE AI system is a developed to predict property prices using machine learning techniques. It analyzes factors such as location, plot size, road width, number of bedrooms, and bathrooms to generate accurate price estimates. The system uses a Linear Regression model and is implemented as a web-based application using Python and Flask. It includes modules like user authentication, sector selection, property search, prediction, and map visualization. Users can input their requirements and receive instant results with suitable property suggestions. The system is fast, efficient, and easy to use, requiring minimal computational resources. It helps reduce manual effort and supports better decision-making in real estate analysis. Overall, the project demonstrates the effective use of artificial intelligence in solving real-world problems.	PO1-3 PO2-3 PO3-3 PO4-3 PO5-3 PO6-3 PO7-3 PO8-3 PO9-3 PO10-3 PO11-3	PSO1 – 3 PSO2 – 3

Note: 1- Low, 2-Medium, 3- High

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LIST OF ABBREVIATIONS

S. NO.	ACRONYM	ABBREVIATION
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	DBMS	Database Management System
4	GUI	Graphical User Interface
5	ML	Machine Learning
6	SQL	Structured Query Language
7	UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 Objective

The objective of this project is to design and develop an intelligent HOUSE PRICE AI system that predicts property prices and recommends suitable properties based on user requirements. In the modern real estate environment, users often face challenges such as price fluctuations, lack of accurate estimation tools, and difficulty in comparing multiple properties effectively. This project aims to overcome these issues by applying machine learning techniques, specifically Linear Regression, to generate reliable and data-driven price predictions. The system considers key features such as area, plot size, road width, number of bedrooms, and number of bathrooms to estimate property values accurately. In addition to prediction, the system also filters and identifies the most relevant properties from the dataset, enabling users to make informed decisions with minimal effort. Furthermore, the project focuses on providing a user-friendly interface that allows easy interaction, ensuring that even non-technical users can efficiently analyze and select properties. Overall, the objective is to combine machine learning and web technologies to create a smart, efficient, and accessible solution for real estate analysis.

1.2 Overview

The EduGuide AI system is a web-based application developed using Python and the Flask framework, integrating machine learning, student profile management, and recommendation techniques to provide a complete career guidance platform. The system is structured into multiple modules including user registration and authentication, profile input, interest and skill collection, recommendation generation, and opportunity display, ensuring a smooth and organized workflow for the user. Students can register and log into the system, enter their educational background, areas of interest, and existing skills, and receive a personalized list of recommended

internships, online courses, skill-development programs, and government training opportunities suited to their profile .Based on the inputs provided, the system applies a machine learning model to analyze the student's data and match it with available opportunities from a curated dataset, ranking results by relevance and suitability. The recommendation output is displayed in a clean and structured format, allowing students to easily browse through suggestions and take informed decisions about their career path. The system also highlights skill gaps and suggests specific courses that can help students strengthen their profiles and improve their chances of securing internships or employment. This combination of profile analysis, intelligent recommendation, and career-oriented guidance makes the EduGuide AI system highly efficient, personalized, and user-friendly, providing a modern and practical solution for student career development.

1.3 Purpose and Importance

The primary purpose of this project is to simplify the career planning process for students and improve their ability to make informed decisions using artificial intelligence techniques. Traditional approaches to career guidance rely heavily on manual searching across multiple platforms, peer suggestions, or generic counselling sessions, which are often time-consuming, inconsistent, and unable to account for individual student profiles and preferences. This project introduces an intelligent approach that automates the process of opportunity discovery and skill-based recommendation, thereby reducing student effort and significantly improving the relevance and accuracy of the guidance provided .The importance of this system lies in its ability to deliver personalized, data-driven insights that help students identify opportunities that are genuinely aligned with their background, interests, and goals. It enables students to avoid irrelevant courses, discover government training programs they may not have been aware of, and systematically build skills that enhance their employability.

1.4 Data Source Description

The system uses a structured dataset containing detailed information about available opportunities, including internship listings, free online courses, government skill-development programs, and associated skill requirements, which are essential for training and operating the recommendation model. Each record in the dataset includes attributes such as opportunity type, domain or field, required skills, eligibility criteria, duration, and platform or provider details, enabling the system to perform accurate and relevant matching against student profiles. The dataset is processed using libraries such as Pandas and NumPy to ensure data consistency, accuracy, and proper formatting before being used for recommendation generation. Various preprocessing techniques are applied, including data cleaning, handling of missing values, removal of duplicate entries, and encoding of categorical variables into numerical form, as machine learning algorithms require numerical input for processing.

1.5 Project Summarization

The EduGuide AI project presents a complete and intelligent solution for student career guidance by integrating machine learning, web development, and structured data management techniques into a single unified platform. The system includes multiple modules such as user authentication, student profile collection, interest and skill input, machine learning-based recommendation generation, and opportunity display, all working together to provide a seamless and efficient user experience. It allows students to receive personalized recommendations for internships, online courses, skill-development programs, and government training opportunities based on their individual profiles, eliminating the need for manual searching across multiple platforms and significantly reducing the time and effort involved in career planning. Overall, the EduGuide AI project demonstrates how artificial intelligence can be effectively applied to real-world educational challenges, providing a smart, practical, and student-friendly solution for career development and opportunity discovery.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction and Traditional Methods

The literature survey provides an overview of existing research and approaches related to career guidance systems, educational recommendation platforms, and student opportunity discovery tools. In earlier times, career counselling and academic guidance were primarily carried out through traditional methods such as one-on-one counselling sessions, printed brochures, college placement cells, and word-of-mouth recommendations from peers and faculty. Career advisors relied on their personal experience, institutional knowledge, and general awareness of available opportunities to guide students, which often resulted in inconsistent, biased, and highly generalized outcomes that did not account for individual student strengths, interests, or skill levels. These traditional approaches lacked the ability to process large volumes of data simultaneously and could not keep pace with the rapidly growing number of online courses, internship platforms, and government training initiatives that have emerged in recent years. Students were often unaware of relevant opportunities simply due to the absence of a centralized and personalized information system.

2.2 Machine Learning Techniques

The literature survey provides an overview of existing research and approaches related to career guidance systems, educational recommendation platforms, and student opportunity discovery tools. In earlier times, career counselling and academic guidance were primarily carried out through traditional methods such as one-on-one counselling sessions, printed brochures, college placement cells, and word-of-mouth recommendations from peers and faculty. Career advisors relied on their personal experience, institutional knowledge, and general awareness of available opportunities to guide students, which often resulted in inconsistent, biased, and highly generalized outcomes that did not account for individual student strengths, interests, or skill level.

2.3 Web-Based Systems and Visualization Techniques

Modern educational guidance and career recommendation applications are increasingly developed as web-based systems to ensure broad accessibility, ease of use, and efficient delivery of information to students across different devices and locations. Technologies such as Python, Flask, HTML, CSS, and JavaScript are commonly used to build interactive platforms where students can register, input their profiles, and receive recommendations through a structured and visually organized interface. These systems typically include features such as user authentication, profile management, dynamic recommendation display, and filtering mechanisms that allow students to refine results based on specific preferences or requirements. In addition to core functionality, several educational platforms have incorporated data visualization techniques to present information in a more meaningful and engaging manner. The integration of machine learning recommendation engines with web-based interfaces and data visualization tools creates a more comprehensive and effective solution, enabling students not only to receive relevant suggestions but also to track their development, explore opportunities visually, and make more informed decisions about their academic and professional journey.

2.4 Limitations of Existing Systems and Summary

Despite the significant progress made in the development of educational recommendation and career guidance systems, several important limitations continue to exist in current approaches that reduce their overall effectiveness and accessibility. Many existing systems rely on static datasets that are not regularly updated, which means that newly available internships, recently launched government programs, and current online courses are often not reflected in the recommendations provided to students. Advanced machine learning models, while capable of delivering highly accurate recommendations, are frequently complex in design, require substantial computational resources, and demand large volumes of training data, making them less suitable for lightweight academic or institutional deployments.

CHAPTER 3

PROJECT METHODOLOGY

This chapter describes the methodology used to design and develop the EduGuide AI system. It explains the overall workflow, system architecture, and the integration of different modules required for student profile analysis and personalized career recommendation. The system is designed using a web-based approach with Flask, combined with machine learning techniques to ensure accurate and relevant suggestions for each student. It also includes modules such as user authentication, profile and skill input, recommendation generation, and opportunity display.

3.1 Proposed Workflow

The proposed workflow of the EduGuide AI system is designed to provide a structured and efficient approach for student profile analysis and personalized opportunity recommendation. The system begins with user authentication, where students register and log in securely using a SQLite database. Once authenticated, students are directed to the profile input module, where they provide details such as their educational background, domain of interest, technical skills, and career goals. These inputs are collected through a simple and intuitive form-based interface and stored for processing. The collected profile data is then passed to the machine learning recommendation module, where the system analyzes the student's information and matches it against a curated dataset of internships, free online courses, skill-development programs, and government training opportunities. The recommendation engine applies the trained model to rank and filter the most relevant opportunities based on the student's profile, ensuring that the suggestions provided are accurate, personalized, and useful. The results are then displayed in a structured and readable format on the output screen, allowing students to browse through recommendations, understand why each opportunity is relevant, and take further action.

3.2 Architectural Overview

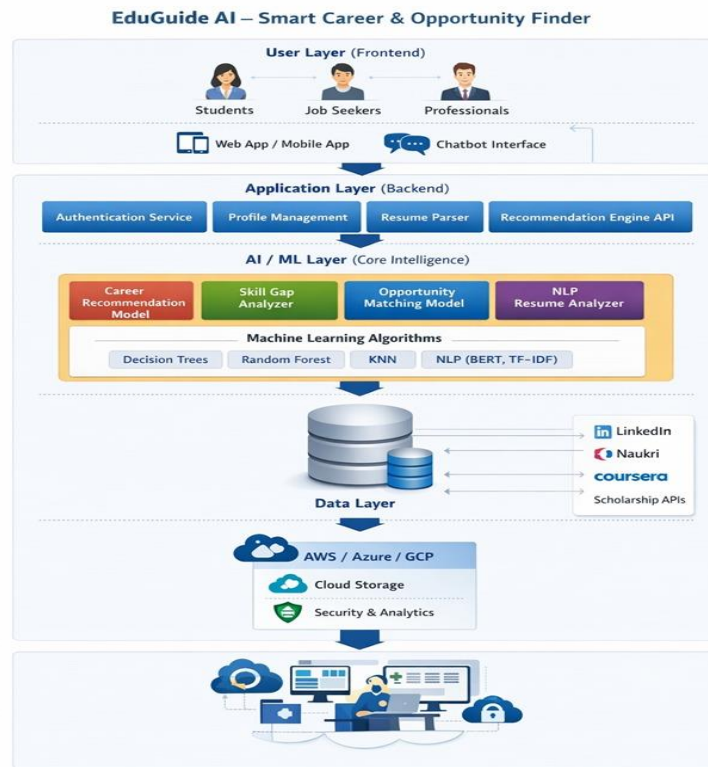


Fig 3.2.1 Architecture Diagram

The architecture of the EduGuide AI system consists of multiple interconnected modules that work together to deliver a seamless and personalized user experience. The central component is the Flask server, which handles all incoming requests and coordinates communication between the various modules of the system. The user authentication module manages student registration and login functionalities using a SQLite database, ensuring that only verified users can access the platform and its features. The student profile module collects and stores key information such as educational background, skills, and areas of interest, which forms the primary input for the recommendation process. The recommendation engine, which serves as the core intelligence of the system, receives the processed profile data and applies the trained machine learning model to generate a ranked list of suitable internships, courses.

3.3 Hardware and Software Requirements

The EduGuide AI system is designed to operate on commonly available hardware with minimal computational requirements, ensuring that it is accessible to students and institutions without the need for specialized or high-end equipment. A computer with at least an Intel i3 processor or equivalent and 4GB of RAM is sufficient to run the application smoothly, although higher specifications can further improve performance and response times during model training and data processing. The software requirements include Python 3.x as the primary programming language, along with supporting libraries such as Flask for web application development, Pandas and NumPy for data processing and manipulation, and Scikit-learn for building and deploying the machine learning recommendation model. The system also uses SQLite as a lightweight and efficient database for storing user credentials and student profile information securely. Front-end components are developed using HTML and CSS to provide a clean and responsive user interface. Development and testing can be carried out using standard tools such as VS Code or PyCharm.

3.4 System Integration and Working

The overall working of the EduGuide AI system involves the seamless integration of multiple components to provide accurate, personalized, and efficient career guidance for students. When a student interacts with the system, the request is received and processed by the Flask server, which routes it to the appropriate module based on the nature of the interaction. The student profile module collects and preprocesses the input data, ensuring it is formatted correctly before being passed to the machine learning recommendation engine. The recommendation engine processes the student's profile features and applies the trained model to generate a ranked list of relevant opportunities, which is then returned to the Flask server and delivered to the front-end interface display.

CHAPTER 4

RELEVANCE OF THE PROJECT

4.1 Why the Model Was Chosen

The EduGuide AI – Smart Career Pathfinder model was chosen because it effectively addresses the challenges faced by students in identifying suitable career opportunities, internships, and skill development programs. In today's competitive environment, students often struggle to find relevant courses and opportunities that match their interests, skills, and academic background. Therefore, a machine learning-based recommendation system is essential to provide accurate and personalized guidance. The system uses a recommendation-based machine learning approach, which is suitable for analyzing student data such as education level, skills, and interests. This model is capable of identifying patterns and matching user profiles with relevant opportunities like internships, free online courses, and government training programs. Compared to manual searching methods, the chosen model significantly reduces time and effort while improving decision-making.

4.2 Comparison with Other Models

Compared to other machine learning models, the EduGuide AI system uses a recommendation-based approach that offers a good balance between performance and complexity. Advanced models such as Deep Learning and Neural Networks can provide highly accurate recommendations, but they require large datasets, high computational power, and complex implementation. Other models like Decision Trees and Random Forest can also be used for classification and recommendation tasks, but they may increase system complexity and reduce interpretability. These models require more processing time and may not be ideal for real-time applications where quick responses are necessary. In contrast, the chosen recommendation model is simple, fast, and easy to implement.

Feature	Linear Regression	Decision Tree	Neural Network
Accuracy	Medium	High	Very High
Speed	Fast	Medium	Slow
Complexity	Low	Medium	High
Data Need	Low	Medium	High
Implementation	Easy	Moderate	Difficult

4.3 Advantages and Disadvantages

The EduGuide AI system offers several advantages that make it highly effective for career guidance and opportunity recommendation. One of the major advantages is its ability to provide personalized recommendations based on individual student profiles. This helps students identify relevant internships, courses, and skill development programs without manually searching across multiple platforms. The system is designed to be user-friendly and accessible, allowing even non-technical users to interact with it. It reduces time consumption and simplifies the decision-making process by presenting filtered and relevant results. The lightweight design ensures that the system runs efficiently on standard hardware without requiring advanced computational resources. Additionally, the modular architecture allows easy expansion, enabling future improvements such as integrating advanced recommendation algorithms and real-time data sources. However, the system also has certain limitations. The accuracy of recommendations depends on the quality and completeness of the input data provided by users. If user data is incomplete or incorrect, the recommendations may not be fully accurate.

4.4 Summary of Relevance

The EduGuide AI – Smart Career Pathfinder project demonstrates the practical importance of machine learning in solving real-world problems related to career planning and skill development. It provides an intelligent and automated solution that helps students discover suitable opportunities based on their profile, thereby improving employability and career growth. By integrating machine learning with a user-friendly interface, the system simplifies the process of searching for internships, courses, and training programs. It reduces manual effort and ensures that users receive relevant and personalized recommendations. The project is especially valuable for students, as it helps them make informed decisions and build stronger resumes. Furthermore, the system serves as a strong foundation for future enhancements such as real-time data integration, advanced recommendation models, and mobile application support. It highlights the role of intelligent systems in modern education and career guidance. Overall, the project proves that technology can significantly improve decision-making processes and support students in achieving their career goals effectively.

CHAPTER 5

MODULE DESCRIPTION

5.1 User Layer Module

The User Layer Module represents the primary interface through which end users interact with the EduGuide AI platform. This module is designed to cater to three distinct categories of users — Students, Job Seekers, and Professionals — each of whom has unique needs and expectations from a smart career and opportunity finder system. Students may use the platform to explore potential career paths, identify skill gaps relative to their academic background, and discover scholarship or internship opportunities. Job seekers, on the other hand, are primarily focused on finding suitable employment opportunities that align with their qualifications, experience, and career aspirations. Professionals may engage with the platform to explore upskilling opportunities, career transitions, or advanced certifications that can accelerate their growth within their respective industries. To serve this diverse user base effectively, the module provides access through two distinct channels — a Web Application and a Mobile Application, as well as a Chatbot Interface. The web and mobile applications offer a full-featured experience with rich visual dashboards, profile management tools, and detailed recommendation displays. The chatbot interface, on the other hand, provides a conversational and intuitive means of interaction, allowing users to ask questions, seek guidance, and receive personalized suggestions in a natural language format. This dual-access approach ensures that the platform is accessible to users regardless of their technical proficiency or preferred mode of interaction, making EduGuide AI inclusive, user-friendly, and adaptable to a wide range of usage contexts and devices.

5.2 Application Layer Module

The Application Layer Module forms the structural backbone of the EduGuide AI system, serving as the intermediary between the user-facing frontend and the intelligent AI/ML core. This layer is responsible for handling all business logic, user

management, and data processing operations that are essential to the platform's functionality. It is composed of four primary services — Authentication Service, Profile Management, Resume Parser, and Recommendation Engine API — each of which plays a specific and critical role in ensuring that the system operates securely, efficiently, and accurately. The Authentication Service is responsible for verifying the identity of users before granting them access to the platform. It manages user registration, login, session handling, and security protocols such as token-based authentication, ensuring that sensitive personal and professional data is accessible only to authorized individuals. The Profile Management service enables users to create, update, and maintain their personal and professional profiles, including educational qualifications, work experience, skills, and career preferences. This profile data serves as the primary input for the AI/ML layer's recommendation algorithms.

5.3 AIML Layer Module

The AI/ML Layer Module is the intellectual core of the EduGuide AI system, housing all the machine learning models and algorithms that drive the platform's smart career guidance and opportunity matching capabilities. This layer is responsible for analyzing user profiles, identifying skill deficiencies, matching users with relevant opportunities, and processing resume content to generate meaningful and personalized recommendations. It is built upon four specialized components — the Career Recommendation Model, the Skill Gap Analyzer, the Opportunity Matching Model, and the NLP Resume Analyzer — each of which addresses a specific dimension of the career guidance process. The Career Recommendation Model uses machine learning techniques to analyze a user's academic background, work experience, and stated interests to suggest suitable career paths and job roles. It leverages historical data and patterns from similar user profiles to generate recommendations that are both realistic and aspirational, helping users make informed decisions about their professional futures. The Skill Gap Analyzer compares a user's current skill set against the requirements of their target career or job role and identifies specific areas where development is needed. This analysis

empowers users to take targeted action by pursuing relevant courses, certifications, or training programs that directly address their gaps.

5.4 Data Layer Module

The Data Layer Module serves as the centralized data management and integration hub of the EduGuide AI platform, responsible for storing, organizing, and retrieving all information that flows through the system. This layer acts as the single source of truth for user profiles, resume data, opportunity listings, recommendation outputs, and interaction histories. A well-structured and efficient data layer is critical to the performance and accuracy of any AI-driven platform, and EduGuide AI's data layer is designed to handle large volumes of structured and unstructured data with reliability and speed. A defining feature of this module is its integration with multiple external data sources that continuously feed fresh and relevant information into the platform. LinkedIn, one of the world's largest professional networking platforms, provides access to job listings, professional profiles, and industry trends. Naukri, a leading job portal in the Indian market, contributes a rich repository of employment opportunities across diverse sectors and experience levels. Coursera, a globally recognized online learning platform, supplies course recommendations and certification programs that can be matched to identified skill gaps. Additionally, Scholarship APIs are integrated to surface funding and educational support opportunities for students who may benefit from financial assistance in pursuing their academic and career goals. By aggregating data from these diverse and authoritative sources, the Data Layer ensures that the recommendations generated by the AI/ML layer are not only personalized but also current, comprehensive, and contextually relevant. The module also manages data pipelines that preprocess, clean, and normalize incoming data before it is stored or consumed by the recommendation models, ensuring data quality and consistency throughout the system. The Data Layer thus plays a foundational role in making EduGuide AI a reliable and continuously updated career guidance ecosystem.

5.5 Cloud Infrastructure Module

The Cloud Infrastructure Module provides the technological foundation upon which the entire EduGuide AI platform is hosted, operated, and scaled. Leveraging the capabilities of leading cloud service providers — Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) — this module ensures that the platform is available, resilient, and performant at all times, regardless of user load or data volume. The choice of a multi-cloud or cloud-agnostic architecture reflects a strategic decision to avoid vendor lock-in, optimize costs, and take advantage of the best-in-class services offered by each provider depending on specific workload requirements. The Cloud Storage component within this module handles the persistent storage of all platform data, including user profiles, uploaded resumes, processed datasets, model artifacts, and recommendation outputs. Cloud storage solutions such as AWS S3, Azure Blob Storage, or Google Cloud Storage provide virtually unlimited, highly durable, and globally accessible data storage that scales automatically with the platform's growth. This eliminates the limitations of on-premises storage infrastructure and ensures that data is always available to the application and AI/ML layers without latency or capacity concerns.

CHAPTER 6

RESULT AND DISCUSSION

6.1 Performance Analysis

The performance analysis of the EduGuide AI – Smart Career Pathfinder system focuses on evaluating its efficiency, recommendation accuracy, response time, and overall system reliability under different usage conditions. The system was tested using multiple user profiles with varying educational backgrounds, skills, and interests to ensure consistent and relevant recommendations across diverse inputs. The machine learning-based recommendation model demonstrated effective performance in suggesting suitable internships, online courses, government training programs, and skill development opportunities based on user data. The system generates recommendations quickly, with response times typically remaining within a few seconds, making it suitable for real-time web-based applications. The use of optimized data processing techniques ensures efficient handling of user input and reduces latency. The integration of backend technologies ensures smooth communication between modules such as user input processing, recommendation engine, and output display. The recommendation accuracy was found to be reliable when sufficient and relevant user data is provided. The system successfully matches user profiles with appropriate opportunities, helping users make informed career decisions. It also reduces the need for manual searching across multiple platforms, thereby saving time and effort. During testing, the system maintained stable performance without crashes or delays, even during repeated usage. It operates efficiently on standard hardware without requiring high computational resources, making it accessible to a wide range of users. However, slight variations in recommendation quality may occur when user input is incomplete or lacks clarity, which highlights the importance of accurate data entry.

6.2 User Feedback

User feedback for the EduGuide AI – Smart Career Pathfinder system was collected from students and users to evaluate its usability, effectiveness, and overall user experience. Most users reported that the system is simple, intuitive, and easy to navigate, with a clear interface that allows them to input their details without confusion. The platform was appreciated for providing a centralized solution where users can access internships, courses, and training programs in one place. The recommendation feature received positive feedback as it provides personalized suggestions based on user profiles, helping students identify relevant opportunities quickly. Users found that the system reduces the time spent searching across multiple websites and helps them focus on suitable options that match their skills and interests. The suggestions related to skill development and resume building were also considered highly useful for improving employability. The system was also appreciated for being beginner-friendly, making it accessible to users with minimal technical knowledge. Most users experienced smooth performance with fast response times and no major technical issues. The structured output of recommendations made it easier for users to understand and compare different opportunities.

However, users suggested several improvements such as adding more advanced filtering options, including real-time updates of opportunities, and expanding the database to include more platforms and resources. Some users also recommended adding features like progress tracking and personalized career roadmaps to enhance functionality. Overall, the feedback indicates that the EduGuide AI system is user-friendly, efficient, and highly useful for students. It has strong potential for further enhancement and can evolve into a comprehensive career guidance platform.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Summary of Outcomes

The EduGuide AI – Smart Career Pathfinder system successfully demonstrates the application of machine learning and web technologies in solving real-world problems related to career guidance and opportunity discovery. The system effectively analyzes user data such as education, skills, and interests to provide personalized recommendations including internships, online courses, government training programs, and skill development suggestions. By integrating multiple modules such as user profile input, recommendation engine, and resume support, the system provides a complete and structured solution for students. During the development and testing phases, the system proved to be efficient, responsive, and reliable, delivering recommendations quickly with minimal computational requirements. The user-friendly interface ensures that both technical and non-technical users can interact with the system easily, while the personalized guidance helps users make informed career decisions. The system significantly reduces the effort required to search for opportunities across multiple platforms and improves the overall career planning process.

The project also highlights the importance of data-driven decision-making in modern education and employment scenarios. By using intelligent recommendation techniques, the system helps students identify suitable opportunities and improve their skills, thereby increasing their employability. Overall, the EduGuide AI system achieves its objectives successfully and serves as a strong foundation for further development and practical implementation.

7.2 Future Scope and Enhancements

Although the EduGuide AI – Smart Career Pathfinder system performs effectively, there are several opportunities for future enhancements that can further improve its functionality, accuracy, and scalability. One of the major improvements is the integration of advanced machine learning models such as collaborative filtering, deep learning, or hybrid recommendation systems, which can provide more accurate and dynamic suggestions. The system can be enhanced by incorporating real-time data from various internship portals, online learning platforms, and government websites, ensuring that users receive up-to-date and relevant opportunities. Additional features such as personalized career roadmaps, progress tracking, and performance analytics can further improve user engagement and provide long-term guidance. The platform can also be expanded to support a wider range of users by including multiple domains, industries, and geographic locations. Enhancements in the user interface, such as interactive dashboards and advanced filtering options, can improve usability and overall user experience. Furthermore, mobile application development and cloud-based deployment can make the system more accessible and scalable. Security and data privacy can also be strengthened by implementing advanced authentication and encryption mechanisms. In addition, integrating features such as chatbot assistance and AI-based career counseling can provide more interactive and intelligent support to users. Overall, the EduGuide AI system has strong potential for future growth and can evolve into a comprehensive and intelligent career guidance platform that supports students in achieving their professional goals.

APPENDICES

APPENDIX A

SOURCE CODE

```
import streamlit as st

import json, os, hashlib, random

# PAGE CONFIG

st.set_page_config(

page_title="EduGuide AI",

page_icon="🔗",

layout="wide",

)

# DATABASE FILE

DB_FILE = os.path.join(os.path.dirname(os.path.abspath(__file__)), "users.json")

def load_db():

if os.path.exists(DB_FILE):

with open(DB_FILE) as f:

return json.load(f)

return {}

def save_db(db):

with open(DB_FILE, "w") as f:

json.dump(db, f, indent=2)

def hash_password(pw):

return hashlib.sha256(pw.encode()).hexdigest()
```

USER AUTHENTICATION

```
def register_user(username, password, email=""):
```

```
    db = load_db()
```

```
    if username in db:
```

```
        return False, "User already exists"
```

```
    db[username] = {
```

```
        "password": hash_password(password),
```

```
        "email": email
```

```
    }
```

```
    save_db(db)
```

```
    return True, "Registered successfully"
```

```
def login_user(username, password):
```

```
    db = load_db()
```

```
    if username in db and db[username]["password"] == hash_password(password):
```

```
        return True, "Login successful"
```

```
    return False, "Invalid credentials"
```

CAREER DATASET

```
CAREERS = {
```

```
    "Software Engineer": {
```

```
        "skills": ["python", "java", "c++", "sql", "algorithms"],
```

```
        "salary": "₹8–30 LPA",
```

```
        "growth": "High"
```

```
    },
```

```
"Data Scientist": {  
  "skills": ["python","machine learning","statistics","pandas"],  
  "salary": "₹10–35 LPA",  
  "growth": "High"  
},
```

```
"Web Developer": {  
  "skills": ["html","css","javascript","react"],  
  "salary": "₹5–20 LPA",  
  "growth": "High"  
},
```

```
"Cybersecurity Analyst": {  
  "skills": ["networking","linux","security","python"],  
  "salary": "₹7–25 LPA",  
  "growth": "High"  
}
```

```
"Web Developer": {  
  "skills": ["html","css","javascript","react"],  
  "salary": "₹5–20 LPA",  
  "growth": "High"  
},
```

```

"UI/UX Designer": {
    "skills": ["figma", "design", "prototyping"],
    "salary": "₹5–18 LPA",
    "growth": "Moderate"
}
}

# MATCHING FUNCTION

def match_careers(user_skills):
    results = []

    for career, info in CAREERS.items():
        required = set(info["skills"])
        user_set = set(user_skills)
        matched = user_set & required
        score = len(matched) / len(required)
        results.append((career, score, matched, required - user_set))

    return sorted(results, key=lambda x: x[1], reverse=True)

# SESSION STATE

if "logged_in" not in st.session_state:
    st.session_state.logged_in = False

# UI

st.title("⚡ EduGuide AI – Career Recommendation System")

menu = st.sidebar.selectbox("Navigation", ["Login", "Register", "Skill Analysis"])

```

```

# REGISTER PAGE

if menu == "Register":

    st.subheader("Register")

    username = st.text_input("Username")

    password = st.text_input("Password", type="password")

    email = st.text_input("Email")

    if st.button("Register"):

        success, msg = register_user(username, password, email)

        if success:

            st.success(msg)

        else:

            st.error(msg)

# LOGIN PAGE

elif menu == "Login":

    st.subheader("Login")

    username = st.text_input("Username")

    password = st.text_input("Password", type="password")

    if st.button("Login"):

        success, msg = login_user(username, password)

        if success:

            st.session_state.logged_in = True

            st.success(msg)

        else:

```

```

st.error(msg)

# SKILL ANALYSIS PAGE

elif menu == "Skill Analysis":

    if not st.session_state.logged_in:

        st.warning("Please login first")

    else:

        st.subheader("Enter Your Skills")

        skills = st.text_area("Enter skills separated by comma")

        if st.button("Analyze"):

            user_skills = [s.strip().lower() for s in skills.split(",")]

            results = match_careers(user_skills)

            st.subheader("Top Career Matches")

            for career, score, matched, missing in results:

                st.write(f"### {career}")

                st.write(f"Match Score: {round(score*100,2)}%")

                st.write(f"Matched Skills: {' '.join(matched)}")

                st.write(f"Missing Skills: {' '.join(missing)}")

            st.write("---")

```

APPENDIX B – Screenshots

The screenshot shows the EduGuide AI dashboard. At the top left is the logo "EDUGUIDE AI" with the tagline "ASGARDIAN CAREER PATHFINDER". The main content area features a large card with a hammer icon and the text "EDUGUIDE AI ASGARDIAN CAREER PATHFINDER". Below this, it says "An AI-powered system that reveals your ideal career path by matching your skills against the wisdom of the nine realms." At the bottom of the card are three buttons: "10 Careers", "53 Skills", and "AI Powered". To the right of the card is a login/register form with tabs for "LOGIN" and "REGISTER". The form includes fields for "WARRIOR NAME *", "EMAIL (optional)", "SECRET RUNE *", and "CONFIRM RUNE *". Below the form are two buttons: "FORGE ACCOUNT" and "Back to Login". At the bottom of the form, it says "* Required fields | Data stored locally on your device."

FIG B.1. DASHBOARD

The screenshot shows the EduGuide AI result page. At the top left is the logo "EDUGUIDE AI" with the tagline "ASGARDIAN CAREER PATHFINDER". At the top right is a user profile icon with the name "ARSHATH PAPPU". Below the logo is a button "Refine Skills". On the right side, there is a "Logout" button. The main content area is titled "YOUR ARSENAL - 1 SKILLS" and shows a list of skills, with "algorithms" selected. Below this is a section titled "TOP CAREER MATCHES". It lists two career matches: "SOFTWARE ENGINEER" and "DATA SCIENTIST". Each match includes a description, a match score, and a "VERY HIGH" rating. The "SOFTWARE ENGINEER" match has a match score of 115 and a description "Forge scalable software systems across the nine realms". The "DATA SCIENTIST" match has a match score of 05 and a description "Read the runes of data to reveal hidden truths."

FIG B.2. RESULT

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